

CSCE 46103/56103 Artificial Intelligence – Fall 2026

Greedy Search, A*, and Heuristics

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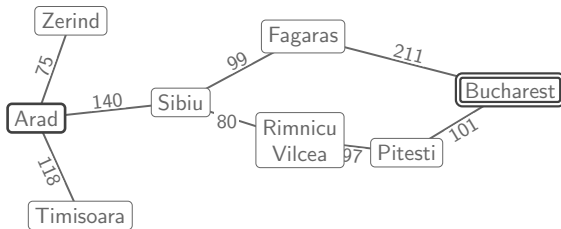
University of Arkansas

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Today's Topics

1. **Announcement of Assignment 1**
2. **Heuristic Estimates:** estimating cost-to-go with $h(n)$
3. **Greedy Best-First Search:** ordering by $h(n)$
4. **A* Search:** ordering by $g(n) + h(n)$
5. **Strategy Comparison:** UCS, Greedy, and A*

Foresight on the Romania Map



Known exactly

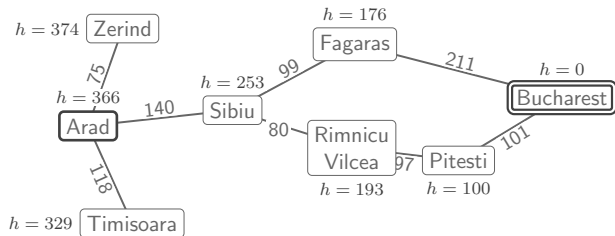
$g(n)$ is the road distance already traveled from Arad to node n .

Estimated ahead

$h(n)$ estimates the remaining distance from node n to Bucharest.

A useful search priority can look forward as well as backward.

Foresight on the Romania Map



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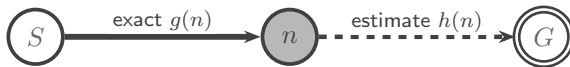
Estimated ahead

$h(n)$ estimates the remaining distance from node n to Bucharest.

A useful search priority can look forward as well as backward.

Heuristic Cost-to-Go

$g(n)$: exact cost from the start to n
 $h(n)$: estimated cost from n to a goal



The graph and path costs do not change; the heuristic changes which frontier node looks promising.

Interpretation

$h(n)$ estimates the cheapest remaining solution cost through n .

Useful conventions

- ▶ same units as path cost,
- ▶ $h(n) \geq 0$,
- ▶ $h(G) = 0$.

One Search, Three Priorities

Strategy	Priority	Exact past	Estimated future	Selection idea
Uniform-cost	$f(n) = g(n)$	✓		cheapest path so far
Greedy best-first	$f(n) = h(n)$		✓	apparently closest to a goal
A*	$f(n) = g(n) + h(n)$	✓	✓	lowest estimated total cost



Greedy Best-First Search

$$f_{\text{Greedy}}(n) = h(n)$$

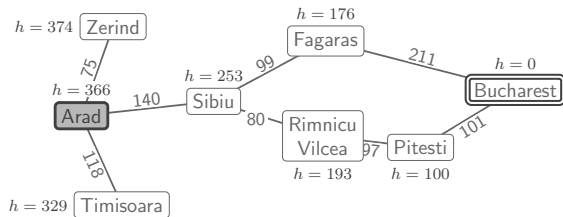
Select the frontier node with minimum $h(n)$.

Uses $h(n)$: estimated remaining cost to a goal.

Ignores $g(n)$: path cost already paid to reach n .

Goal-directed, but not necessarily cost-optimal.

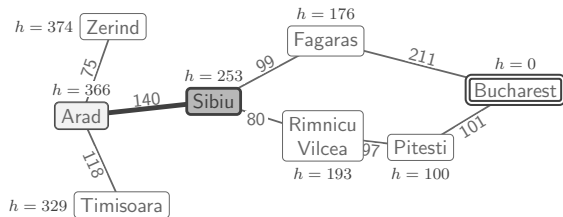
Greedy Romania Trace



Start

Priority: Arad:366

Greedy Romania Trace



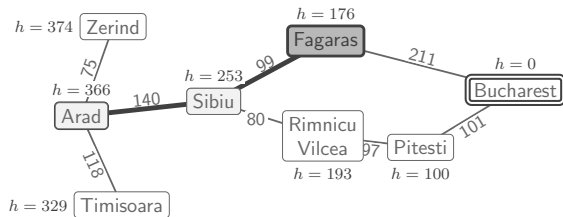
After Arad

Sibiu:253

Zerind:374

Timisoara:329

Greedy Romania Trace



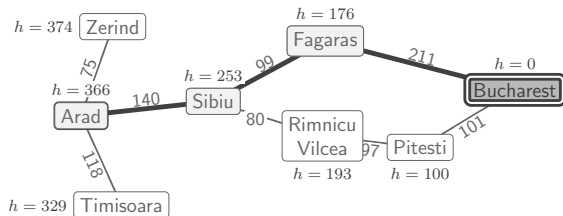
Lowest values after Sibiu

Fagaras:176

Rimnicu:193

Timisoara:329

Greedy Romania Trace



Minimum after Fagaras

Bucharest:0

Returned route

Arad → Sibiu → Fagaras → Bucharest

$$g = 140 + 99 + 211 = 450$$

Greedy Route Cost

Greedy route

Arad $\rightarrow$ Sibiu $\rightarrow$ Fagaras $\rightarrow$ Bucharest

depth = 3, $g = 450$

Lower-cost route

Arad $\rightarrow$ Sibiu $\rightarrow$ Rimnicu Vilcea $\rightarrow$ Pitesti $\rightarrow$
Bucharest

depth = 4, $g = 418$

Fagaras

$g = 239, h = 176$

Rimnicu Vilcea

$g = 220, h = 193$

Greedy chooses the smaller h , even though that node already has the larger g .

Greedy Tradeoffs

Property	Greedy best-first search
Selection	minimum $h(n)$
Complete	finite graphs with reached-state checking; not in general
Cost-optimal	no
Best behavior	a useful heuristic points quickly toward a goal
Worst behavior	a misleading heuristic explores a long or costly detour

What Greedy remembers

A graph-search version records reached states to avoid cycles and repeated exploration.

What Greedy omits

The accumulated cost $g(n)$ does not affect priority.

In the worst case, time and memory remain exponential in the search depth.

Path Cost and Direction

Frontier node	$g(n)$	$h(n)$	$g(n) + h(n)$
Fagaras	239	176	415
Rimnicu Vilcea	220	193	413

Greedy choice

Fagaras has the smaller estimate to the goal:
 $176 < 193$.

Estimated-total choice

Rimnicu Vilcea has the smaller projected route cost: $413 < 415$.

Keep the goal direction, but account for the cost already paid.

A* Search

$$f_{A^*}(n) = g(n) + h(n)$$

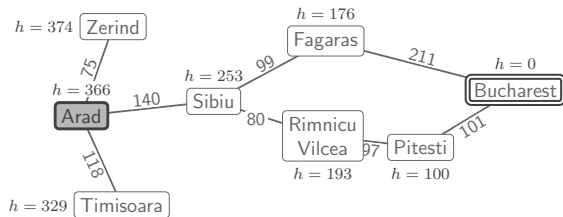
Select the frontier node with minimum $f(n)$.

Uses $g(n)$: exact path cost so far.

Uses $h(n)$: estimated remaining cost to a goal.

Cost-aware and goal-directed.

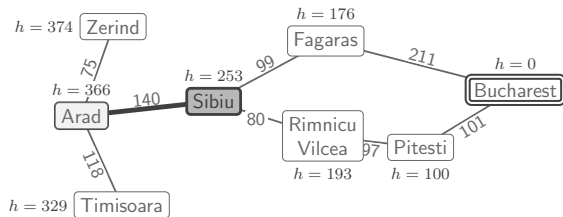
A* Romania Trace I



Start

$$\text{Arad:0} + 366 = 366$$

A* Romania Trace I



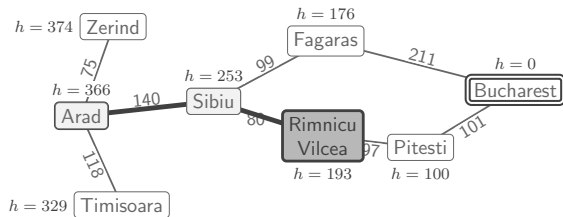
After Arad

$$\text{Sibiu: } 140 + 253 = 393$$

$$\text{Timisoara: } 118 + 329 = 447$$

$$\text{Zerind: } 75 + 374 = 449$$

A* Romania Trace I



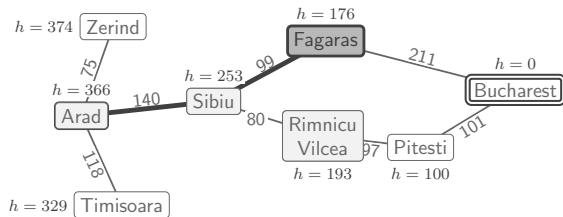
Lowest values after Sibiu

$$\text{Rimnicu: } 220 + 193 = 413$$

$$\text{Fagaras: } 239 + 176 = 415$$

$$\text{Timisoara: } 447$$

A* Romania Trace II



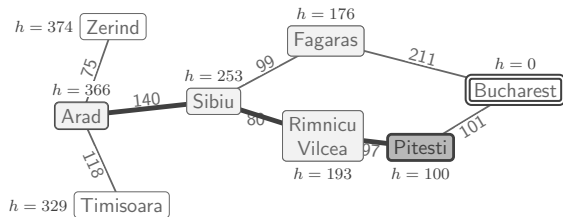
Next minimum f

Fagaras:415

Pitesti:417

Fagaras generates Bucharest with
 $g = f = 450$.

A* Romania Trace II



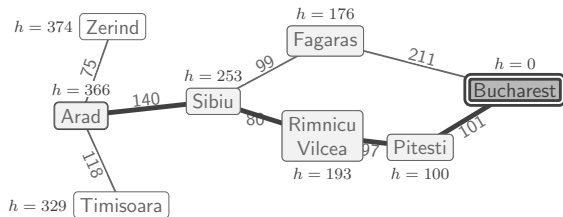
Lower frontier priority

Pitesti: $317 + 100 = 417$

Bucharest: 450

Pitesti generates Bucharest with a cheaper $g = 418$.

A* Romania Trace II



Returned route

Arad → Sibiu → Rimnicu Vilcea → Pitesti
→ Bucharest

$$g = 140 + 80 + 97 + 101 = 418$$

A* and Uniform-Cost Search

$$\begin{aligned} h(n) = 0 \text{ for every node} &\implies f_{A^*}(n) = g(n), \\ &\implies A^* = \text{UCS}. \end{aligned}$$

No guidance

A* falls back to the expanding cost contour of UCS.

Useful guidance

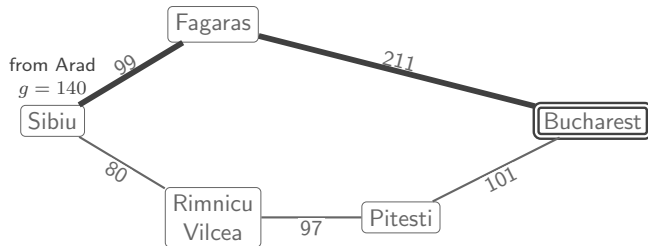
The same best-first framework can select fewer irrelevant nodes.

Unsafe guidance

An overconfident estimate can sacrifice the optimality guarantee.

The heuristic changes the ordering; $g(n)$ remains the actual path cost.

Goal-Test Timing



Goal generated

Fagaras proposes Bucharest:

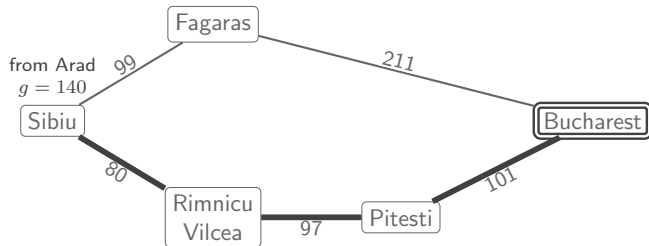
$$g = f = 450.$$

Smaller frontier value

Pitesti remains waiting with $f = 417$.

A generated goal is only a frontier candidate;
return when the goal has minimum priority and is selected.

Goal-Test Timing



Cheaper goal path

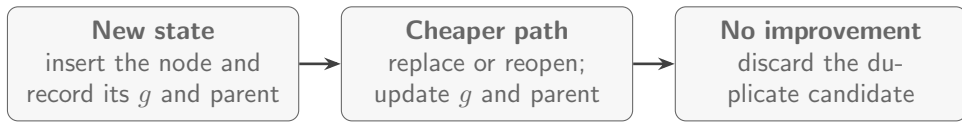
Pitesti proposes Bucharest:

$$g = f = 418.$$

Return on selection.

A generated goal is only a frontier candidate;
return when the goal has minimum priority and is selected.

Best-Cost Records



A* record

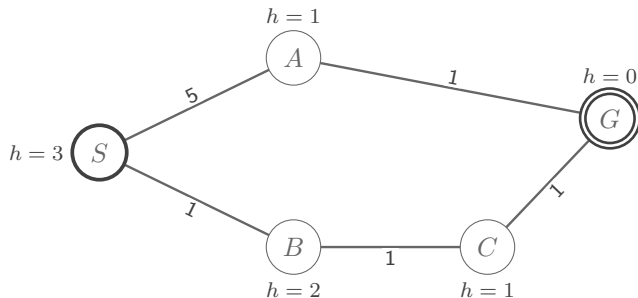
Store the best known g for every reached state.
A lower g changes both the parent and
 $f = g + h$.

Greedy record

In this deck's graph-search convention, mark a state reached when generated because g does not determine priority.

Frontier replacement is necessary even when no expanded state needs reopening.

Five-Node Challenge



Trace rules

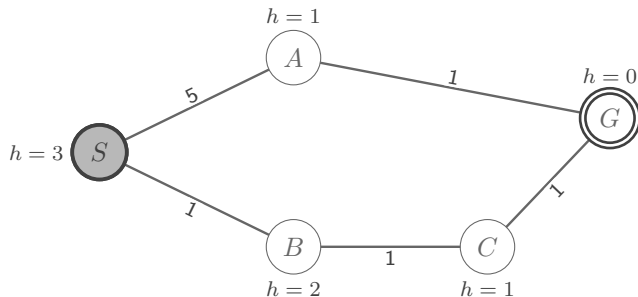
- Greedy selects minimum h .
- A* selects minimum $g + h$.
- Test the goal on selection.

Record for each strategy

Selection order, returned path, and total cost.

Does knowing the exact remaining cost make Greedy cost-optimal?

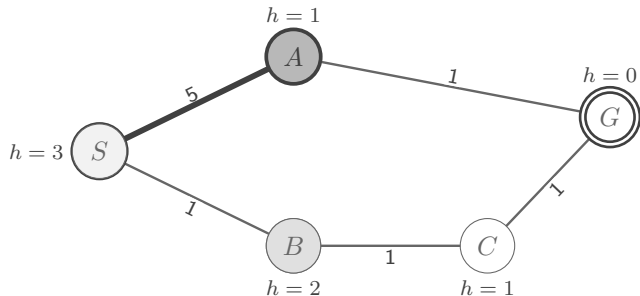
Five-Node Greedy Trace



Select S

$S : h = 3$

Five-Node Greedy Trace



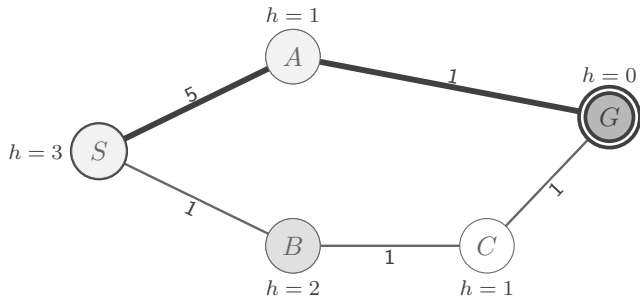
After S

$A : h = 1$

$B : h = 2$

Greedy ignores $g(A) = 5$.

Five-Node Greedy Trace



After A

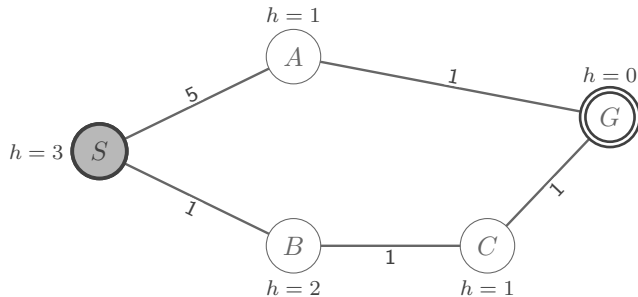
$G : h = 0$

$B : h = 2$

Returned path

$S \rightarrow A \rightarrow G, \quad g = 6.$

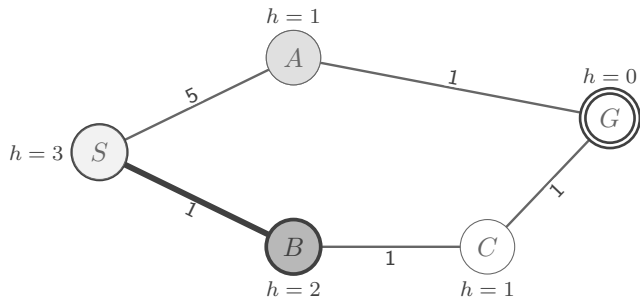
Five-Node A* Trace



Select S

$$S : 0 + 3 = 3$$

Five-Node A* Trace

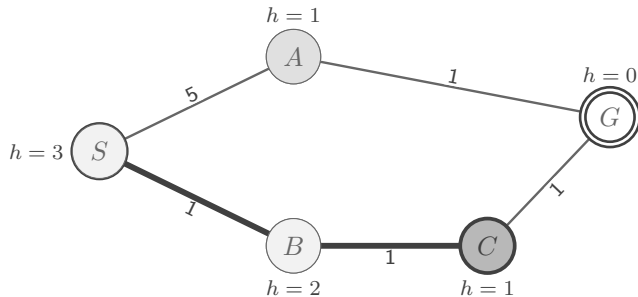


After S

$$B : 1 + 2 = 3$$

$$A : 5 + 1 = 6$$

Five-Node A* Trace

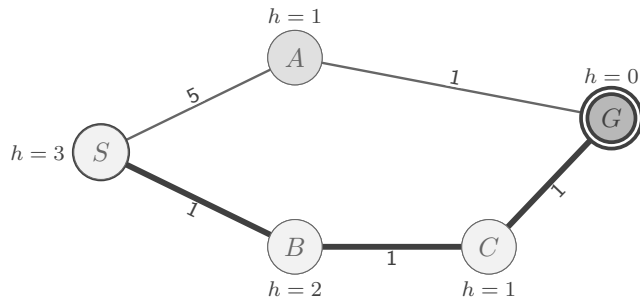


After B

$C : 2 + 1 = 3$

$A : 6$

Five-Node A* Trace



After C

$$G : 3 + 0 = 3$$

$$A : 6$$

Returned path

$$S \rightarrow B \rightarrow C \rightarrow G, \quad g = 3.$$

Three Policies, One Graph

Strategy	Priority	State record	Selection order	Returned path	Cost
UCS	g	best g	S, B, C, G	$S \rightarrow B \rightarrow C \rightarrow G$	3
Greedy	h	reached	S, A, G	$S \rightarrow A \rightarrow G$	6
A*	$g + h$	best g	S, B, C, G	$S \rightarrow B \rightarrow C \rightarrow G$	3

Perfect estimate

On this graph, every $h(n)$ equals the true cheapest remaining cost.

Greedy failure

Perfect knowledge of the future is insufficient when the priority ignores the expensive past.

A* Guarantees

Admissible estimate

Never overestimate the true cheapest remaining cost:

$$h(n) \leq h^*(n).$$

Consistent estimate

Respect each step-cost triangle:

$$h(n) \leq c(n, n') + h(n').$$

High-level result

With finite branching, $c(n, n') \geq \epsilon > 0$, and admissible h , A* is complete. With best- g records and reopening when a better path is found, it returns a lowest-cost solution; consistency makes those reopens unnecessary.

The next deck develops these conditions, their proofs, and heuristic construction.

Heuristic Failure Modes

Heuristic behavior	Greedy best-first	A*
Weak, admissible, and consistent	Weak or misleading direction; a costly path may win.	Still cost-optimal, but may expand many nodes; $h = 0$ gives UCS.
Admissible but inconsistent	No additional guarantee; priority still uses only h .	Best- g updates and reopening may be required.
Overestimating and inadmissible	Can chase the wrong region and has no cost guarantee.	May return a suboptimal solution; the guarantee is lost.

Accuracy question

How close is $h(n)$ to $h^*(n)$? This changes search order and the number of expansions.

Safety question

Does h avoid overestimation and respect step-cost triangles? This controls A*'s guarantee and reopening.

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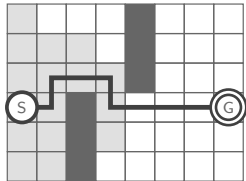
Does h avoid overestimation and respect step-cost triangles? This controls A*'s guarantee and reopening.

Even exact h^* does not make Greedy cost-optimal.

For A*: accuracy affects work; admissibility and consistency affect guarantees.

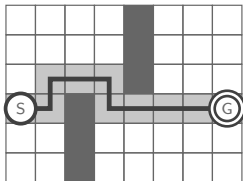
Grid Search Patterns

UCS: cost wavefront



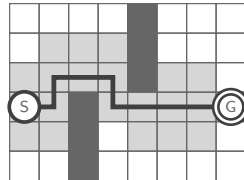
Expands outward by accumulated cost.

Greedy: goal pull



Concentrates where h decreases.

A*: focused wavefront



Balances path cost and goal direction.

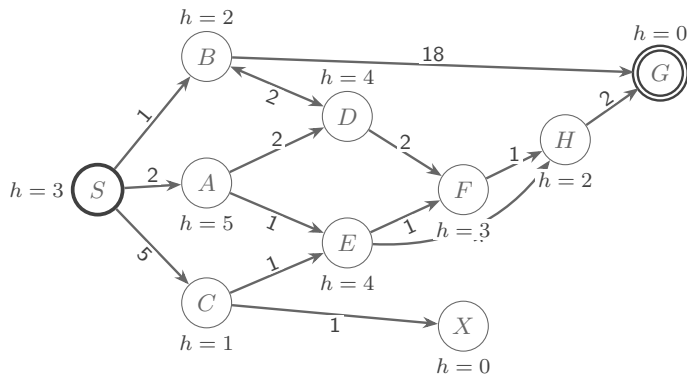
Schematic: the heuristic reshapes the explored region; obstacles and tie-breaking still matter.

Strategy Selection

Situation	Preferred starting point	Reason
No useful heuristic; lowest cost required	Uniform-cost search	Uses exact path cost and preserves the cost guarantee.
Useful heuristic; lowest cost required	A*	Adds goal direction while retaining the guarantee under the stated conditions.
Speed matters more than path cost	Greedy best-first	Uses the heuristic alone and may reach a goal with fewer selections.
Weak heuristic or tight memory	Revisit the formulation or method	A* can inherit UCS-like exploration and a large frontier.

The best strategy depends on the guarantee required and the information available.

Informed-Search Challenge



Successor order

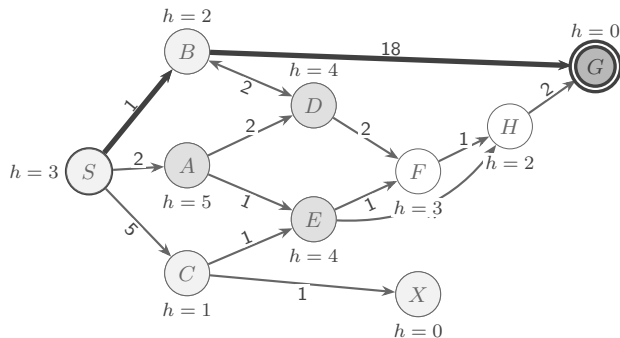
S	A, B, C
A	D, E
B	D, G
C	X, E
D	F, B
E	F, H
F	H
H	G

Trace conventions

Test goals on selection. Greedy marks reached states on generation. A* keeps the best g and replaces a strictly worse frontier record. Break priority ties by earlier insertion.

For Greedy and A*: trace the priority frontier; record the selection order, returned path, depth, cost, peak frontier, and replacements.

Greedy Challenge



Select	<i>h</i> -frontier afterward
<i>S</i>	[<i>C</i> : 1, <i>B</i> : 2, <i>A</i> : 5]
<i>C</i>	[<i>X</i> : 0, <i>B</i> : 2, <i>E</i> : 4, <i>A</i> : 5]
<i>X</i>	[<i>B</i> : 2, <i>E</i> : 4, <i>A</i> : 5]
<i>B</i>	[<i>G</i> : 0, <i>E</i> : 4, <i>D</i> : 4, <i>A</i> : 5]
<i>G</i>	return

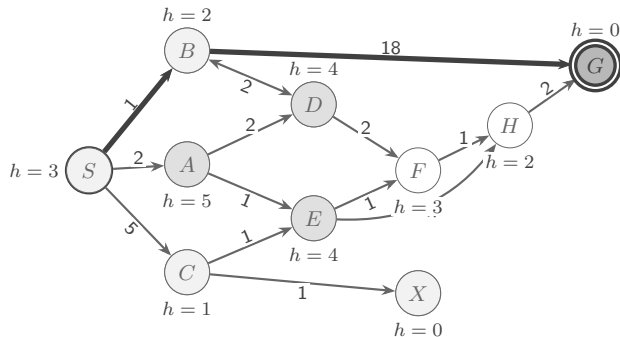
Returned path

$S \rightarrow B \rightarrow G$

depth = 2, $g = 19$

Peak frontier size: 4.

Greedy Challenge



Select	h -frontier afterward
S	$[C : 1, B : 2, A : 5]$
C	$[X : 0, B : 2, E : 4, A : 5]$
X	$[B : 2, E : 4, A : 5]$
B	$[G : 0, E : 4, D : 4, A : 5]$
G	return

Returned path

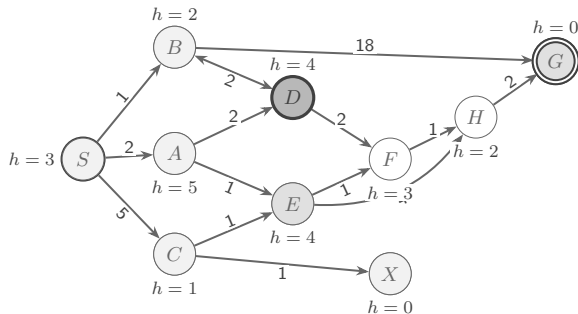
$S \rightarrow B \rightarrow G$

depth = 2, $g = 19$

Peak frontier size: 4.

Greedy explores dead-end X , then accepts the costly $B \rightarrow G$ edge because g never enters its priority.

A* Challenge I



Select	<i>f</i> -frontier afterward
<i>S</i>	[<i>B</i> : 3, <i>C</i> : 6, <i>A</i> : 7]
<i>B</i>	[<i>C</i> : 6, <i>A</i> : 7, <i>D</i> : 7, <i>G</i> : 19]
<i>C</i>	[<i>X</i> : 6, <i>A</i> : 7, <i>D</i> : 7, <i>E</i> : 10, <i>G</i> : 19]
<i>X</i>	[<i>A</i> : 7, <i>D</i> : 7, <i>E</i> : 10, <i>G</i> : 19]
<i>A</i>	[<i>D</i> : 7, <i>E</i> : 7, <i>G</i> : 19]

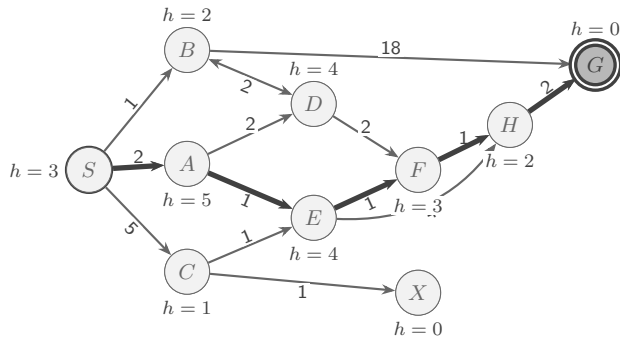
First replacement

A improves *E* from $(g, f) = (6, 10)$ to $(3, 7)$. Its path to *D* has $g = 4$, so the existing $g = 3$ record survives.

Peak frontier size reaches **5** after expanding *C*.

The $f = 7$ ties select *A* before *D*, then *D* before *E*, by earlier insertion.

A* Challenge II



Select	f -frontier afterward
<i>D</i>	[<i>E</i> : 7, <i>F</i> : 8, <i>G</i> : 19]
<i>E</i>	[<i>F</i> : 7, <i>H</i> : 9, <i>G</i> : 19]
<i>F</i>	[<i>H</i> : 7, <i>G</i> : 19]
<i>H</i>	[<i>G</i> : 7]
<i>G</i>	return

Four replacements

$E : 10 \rightarrow 7, \quad F : 8 \rightarrow 7,$

$H : 9 \rightarrow 7, \quad G : 19 \rightarrow 7.$

The first goal record waits until it has minimum f .

Returned path: $S \rightarrow A \rightarrow E \rightarrow F \rightarrow H \rightarrow G$
 depth 5, cost **7**, peak frontier **5**.

Challenge Debrief

Strategy	Selection order	Returned path	Depth	Cost	Peak	Repl.
Greedy	S, C, X, B, G	$S \rightarrow B \rightarrow G$	2	19	4	0
A*	$S, B, C, X, A,$ D, E, F, H, G	$S \rightarrow A \rightarrow E \rightarrow F \rightarrow$ $H \rightarrow G$	5	7	5	4

Challenge Debrief

Strategy	Selection order	Returned path	Depth	Cost	Peak	Repl.
Greedy	S, C, X, B, G	$S \rightarrow B \rightarrow G$	2	19	4	0
A*	$S, B, C, X, A,$ D, E, F, H, G	$S \rightarrow A \rightarrow E \rightarrow F \rightarrow$ $H \rightarrow G$	5	7	5	4

Greedy trap

Low h pulls the search through C and dead-end X ; the costly past through B is invisible.

A* recovery

Best- g records repeatedly lower f until the cost-7 goal record becomes selectable.

Safe, not perfect

This h is admissible and consistent. Still, $h(X) = 0$ does not make X a goal or useful.

A safe heuristic preserves A*'s guarantee; it does not guarantee a perfect search order.

Informed-Search Takeaways

Uniform-cost search

lowest g first

$$f(n) = g(n)$$

Exact past cost; no goal estimate.

Greedy search

lowest h first

$$f(n) = h(n)$$

Strong direction; no cost guarantee.

A* search

lowest $g + h$ first

$$f(n) = g(n) + h(n)$$

Balances cost and direction.

Today

Use heuristic guidance

Next

Design and verify h

Then

Optimize complete states

Thank You

Any questions?

References

S. J. Russell and P. Norvig. *Artificial Intelligence: A Modern Approach*. Pearson, 4th edition, 2020. URL <https://aima.cs.berkeley.edu/>.